WORKED SOLUTION

STEP III Specimen paper, Question 12

Mechanics • Equilibrium of a Rod

STEP III Specimen paper, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

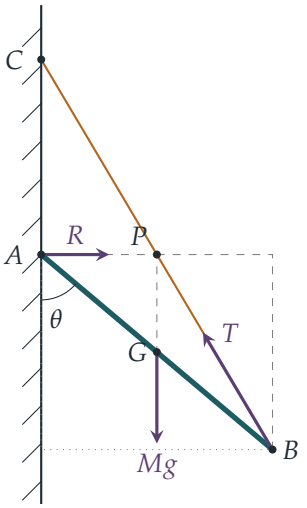
step.maths.org/questions/spec-s3-q12

The key idea

The rod is acted on by exactly three forces: its weight, the reaction from the smooth wall and the tension in the string. Three non-parallel forces holding a body in equilibrium must pass through a single point, and here that point is easy to locate. It pins down the geometry completely, and a single resolution then gives the tension.

Solution

Model. The rod is uniform, so its weight Mg acts at its midpoint G. The wall is smooth, so the reaction R at A is horizontal (perpendicular to the wall). The string is light and inextensible, so it pulls on B with a tension T directed along BC.



The inclination of the rod

The reaction R acts along the horizontal line through A, and the weight acts along the vertical line through G. These meet at the point P level with A and directly above (or below) G. Since the three forces are in equilibrium, the moments about P must balance; R and Mg have no moment about P, so T has none either, and the line of the string must pass through P.

Ask yourself Does the rod go up or down from A?

Measure horizontal distances from the wall. B is at distance $L\sin\theta$ from the wall, where θ is the angle between the rod and the vertical, and G (hence P) is at half that distance, $\frac{1}{2}L\sin\theta$. The string runs in a straight line from B to C, which is on the wall, so a point halfway across horizontally is halfway along the string: P is the *midpoint* of BC. So the height of P is the average of the heights of B and C. But P is level with A and C is above A, so B must be *below* A, by exactly the distance AC.

So the rod hangs downwards from A. Let θ be its angle with the downward vertical. Then B is a depth $L\cos\theta$ below A, and so $AC = L\cos\theta$. Relative to B, the point C is a horizontal distance $L\sin\theta$ away and a height $L\cos\theta + L\cos\theta = 2L\cos\theta$ above. Since the string has length αL, Pythagoras gives

$$\alpha^2 L^2 = L^2\sin^2\theta + 4L^2\cos^2\theta = L^2\left(1 + 3\cos^2\theta\right),$$

so

$$\cos^2\theta = \frac{\alpha^2 - 1}{3}, \qquad \text{i.e.} \qquad \theta = \cos^{-1}\sqrt{\frac{\alpha^2-1}{3}},$$

with the rod pointing *downwards* from A at this angle to the vertical.

The possible values of α

We need C strictly above A, so $\cos\theta > 0$. (If $\cos\theta = 0$ then C coincides with A and the rod is horizontal; then A, B, C are collinear.) We also need A, B, C not collinear, so the rod is not vertical: $\sin\theta \neq 0$, i.e. $\cos^2\theta < 1$. Hence

$$0 < \frac{\alpha^2 - 1}{3} < 1 \iff 1 < \alpha^2 < 4 \iff 1 < \alpha < 2,$$

using $\alpha > 0$. Conversely, for any α in this range the angle above exists and the construction gives a configuration in which the three forces are concurrent, so moments balance, and the resolutions below give a positive T and R. The set of possible values is $1 < \alpha < 2$.

The tension

Resolve vertically. The string makes an angle with the vertical whose cosine is (vertical rise)/(length) $= 2L\cos\theta/(\alpha L)$, and R is horizontal, so

$$T \cdot \frac{2\cos\theta}{\alpha} = Mg \quad \Longrightarrow \quad T = \frac{Mg\alpha}{2\cos\theta} = \frac{Mg\alpha}{2}\left(\frac{3}{\alpha^2-1}\right)^{\frac{1}{2}},$$

substituting $\cos\theta = \sqrt{(\alpha^2-1)/3}$ (the positive root, since $\cos\theta > 0$). This is the required result.

For completeness, resolving horizontally gives $R = T\cdot\dfrac{L\sin\theta}{\alpha L} = \frac{1}{2}Mg\tan\theta > 0$, so the wall does push outwards on the rod, as a wall must.

Checks

- As $\alpha \to 2^-$, $\cos\theta \to 1$: the rod hangs vertically below A with the string doubled back along it, and $T \to Mg$, since the string then carries the whole weight. As $\alpha \to 1^+$, $\theta \to 90^\circ$ and $T \to \infty$: a nearly horizontal rod needs a huge tension to hold it up.
- With $\alpha = 1.5$, we get $\cos\theta \approx 0.6455$ ($\theta \approx 49.8^\circ$) and $T \approx 1.162\,Mg$. We checked numerically, for thousands of random α in $(1,2)$, that the moment about A vanishes at exactly one angle, which matches the formula, along with the tension; and that no equilibrium exists for α outside $(1,2)$.

Common slips

- Drawing the rod sloping *upwards* from A, as in a ladder problem. The concurrency argument forces B below A, and with the wrong picture the equations have no consistent solution.
- Forgetting the reaction at a smooth wall is horizontal, and so introducing an unnecessary friction or vertical component.
- Including $\alpha = 1$ or $\alpha = 2$: these give collinear (degenerate) configurations, which the question excludes.

Beyond STEP

The three-force principle is a special case of a general fact in statics: a system of forces is in equilibrium exactly when its resultant force and its resultant moment about any one point both vanish. Choosing the point cleverly, as we did with P, is often the quickest route. The same idea underlies the analysis of trusses, cables and frameworks in engineering mechanics. For a thorough treatment of forces and torques, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.